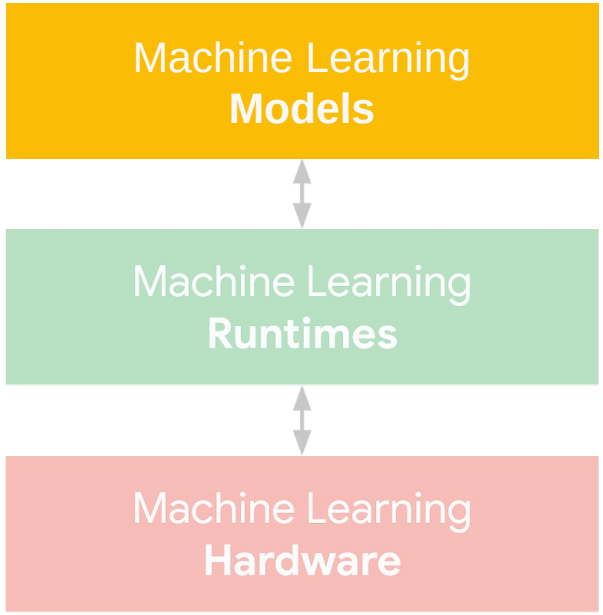
What are the Challenges for TinyML?

Part D



Half-screen. Show presenter.

Model Compression Techniques

Pruning

Quantization

Knowledge Distillation

...

Half-screen. Show presenter.

Model Compression Techniques

Pruning

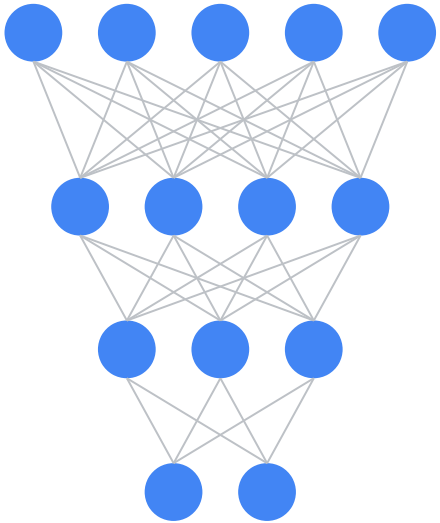
Quantization

Knowledge Distillation

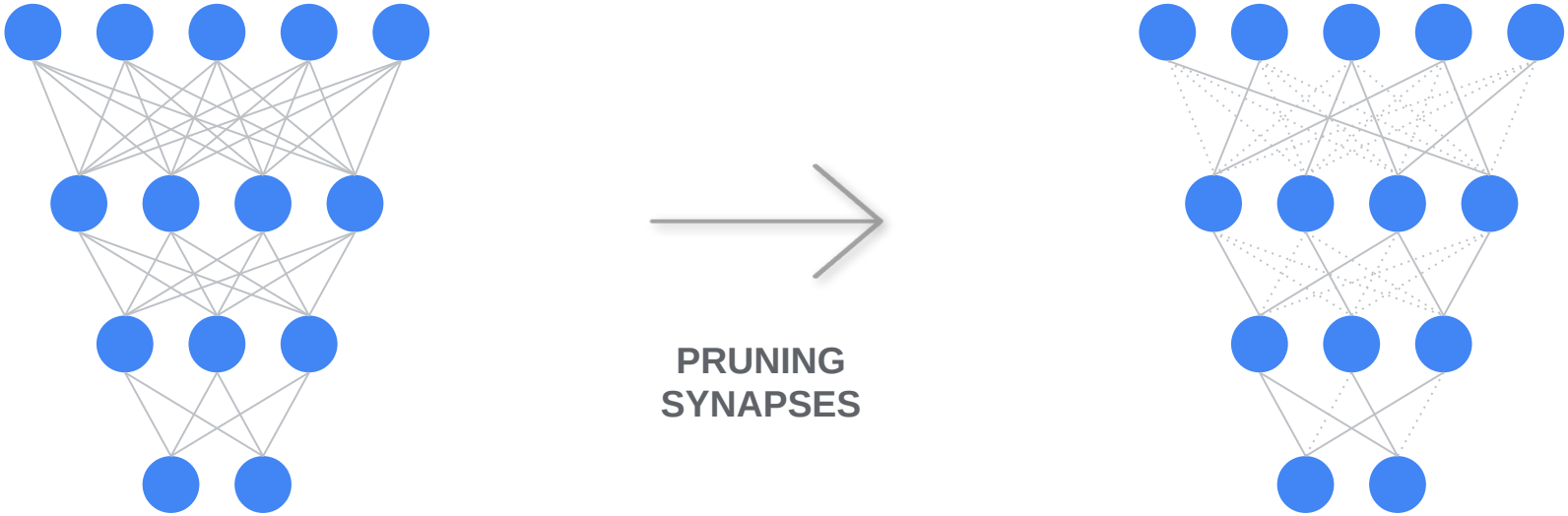
...

Half-screen. Show presenter.

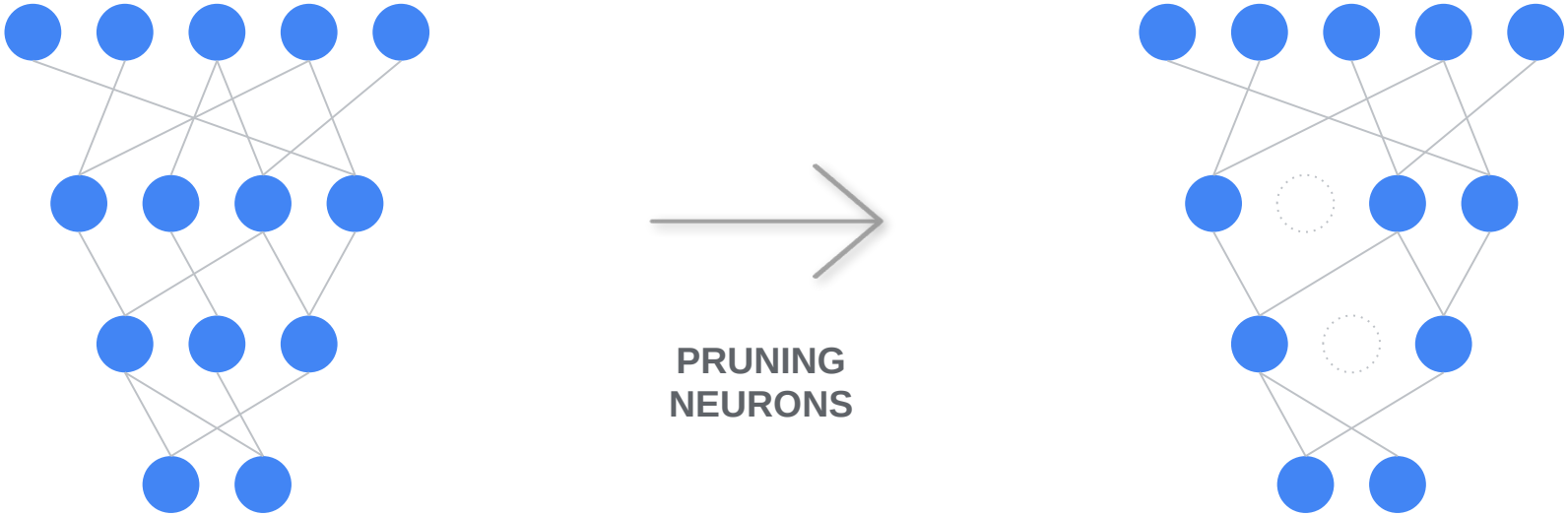
Pruning



Pruning



Pruning



Model Compression Techniques

Pruning

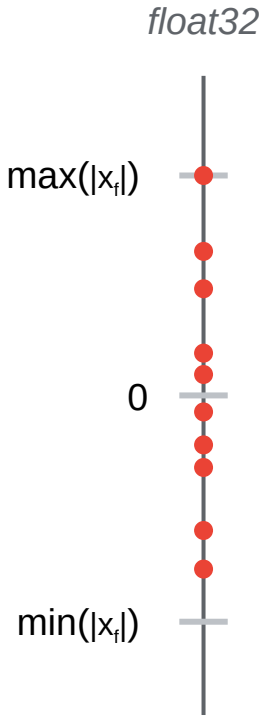
Quantization

Knowledge Distillation

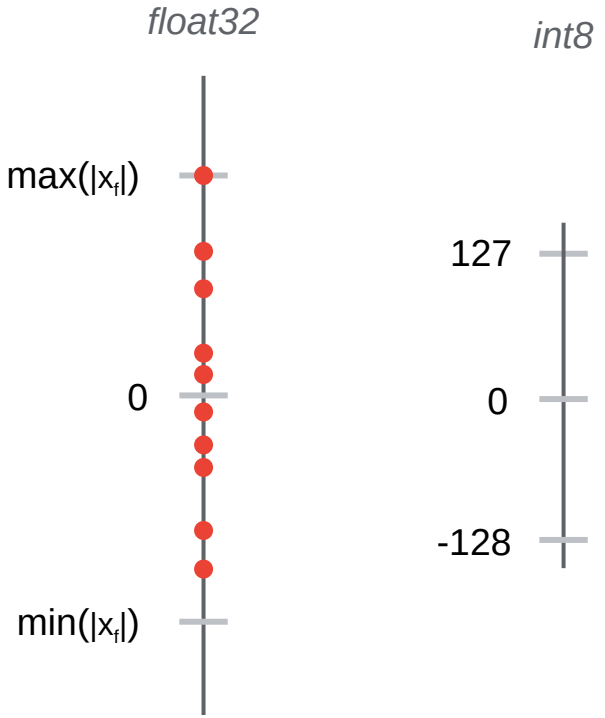
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Half-screen. Show presenter.

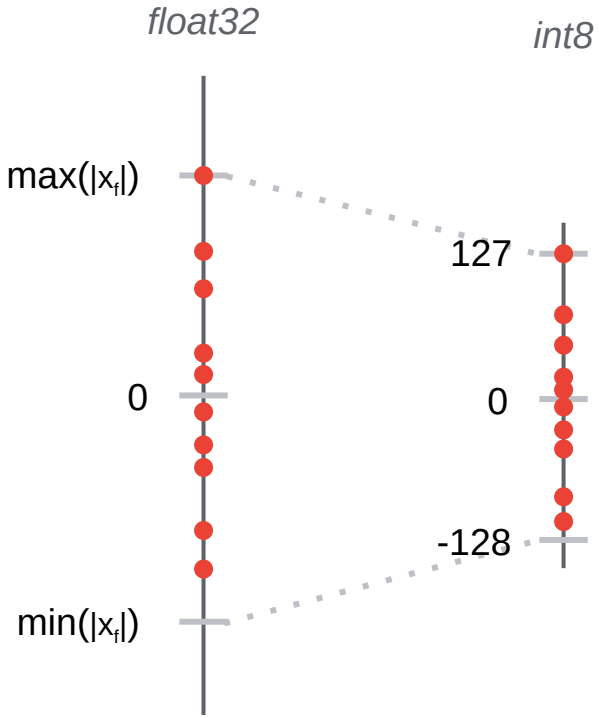
Quantization



Quantization



Quantization



Model Compression Techniques

Pruning

Quantization

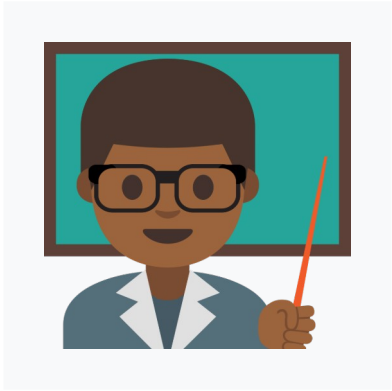
Knowledge Distillation

...

Half-screen. Show presenter.

Knowledge Distillation

TEACHER





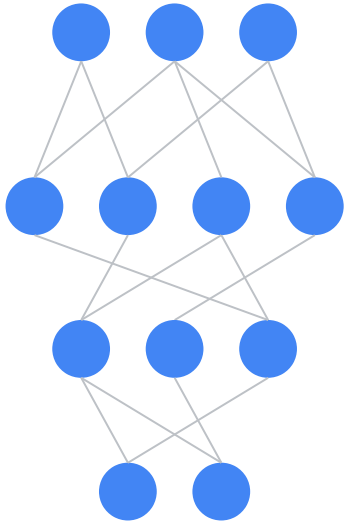
STUDENT



Knowledge Distillation

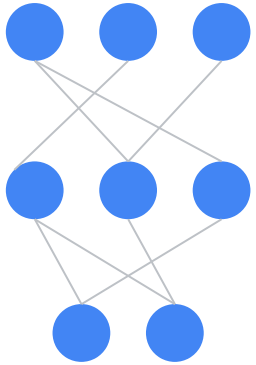
TEACHER

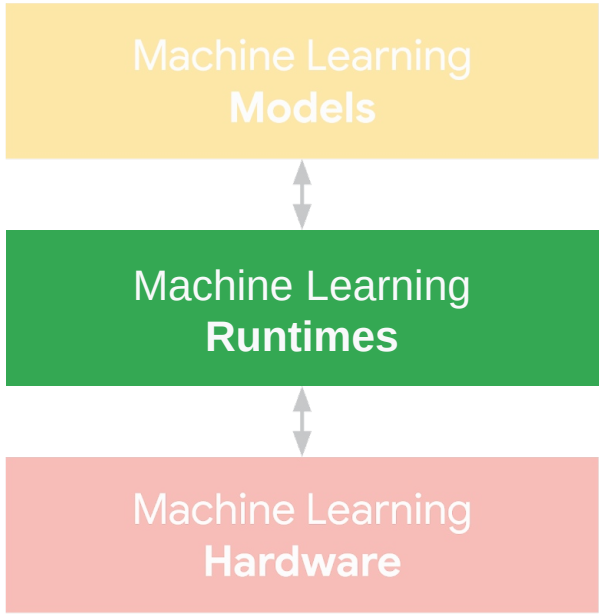




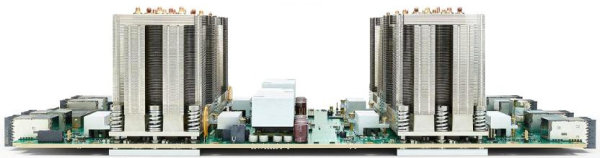
STUDENT

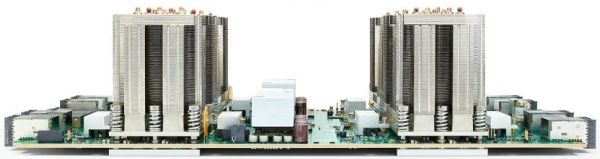






Half-screen. Show presenter.





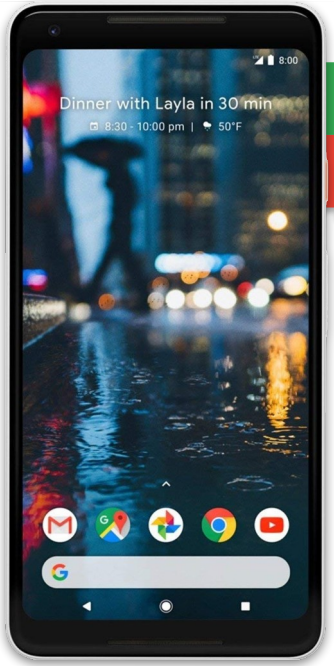


TensorFlow


8:00
Dinner with Layla in 30 min
8:30 - 10:00 pm | 50°F

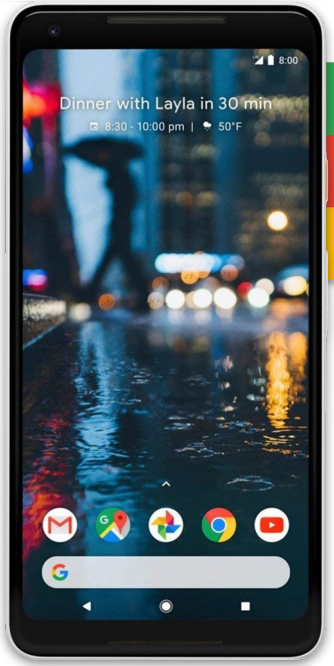


Less memory



Less memory

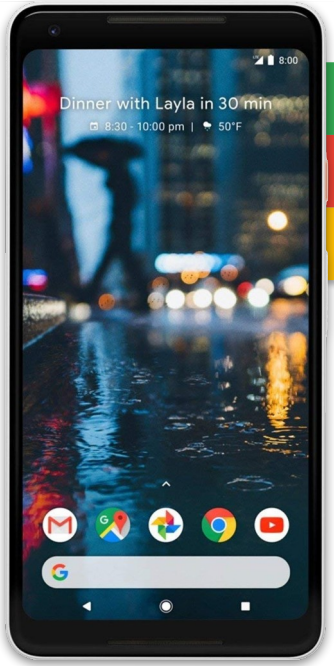
Less compute power



Less memory

Less compute power

Only focused on ***inference***



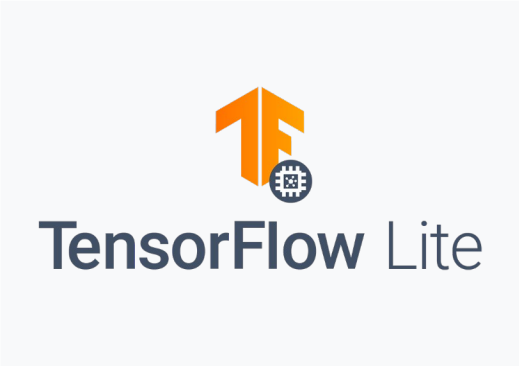
Less memory

Less compute power

Only focused on *inference*

TensorFlow Lite


TensorFlow


TensorFlow Lite

Key Differences

	TensorFlow	TensorFlow Lite
Topology	Variable	Fixed
Weights	Variable	Fixed
Binary Size	Unimportant	High Priority
Distributed Compute	Needed	Not Needed
Developer Background	**ML Researcher**	**Application Developer**

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Architecture



Trained TensorFlow Model



Architecture

Trained TensorFlow Model

↓

TensorFlow Lite **Converter**



Architecture

Trained TensorFlow Model

↓

TensorFlow Lite **Converter**

↓

TensorFlow Lite **Model File** *(.tflite)*



Architecture

Trained TensorFlow Model

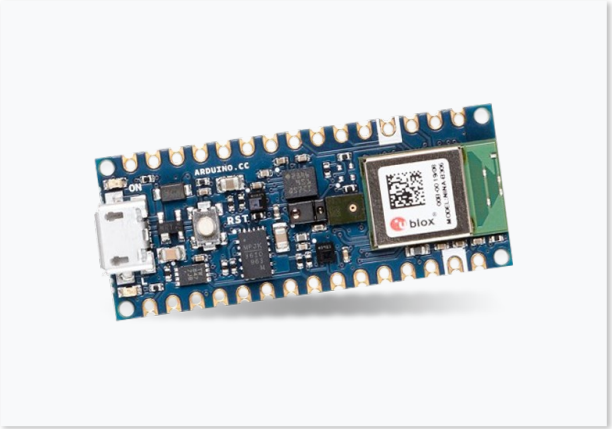
↓

TensorFlow Lite **Converter**

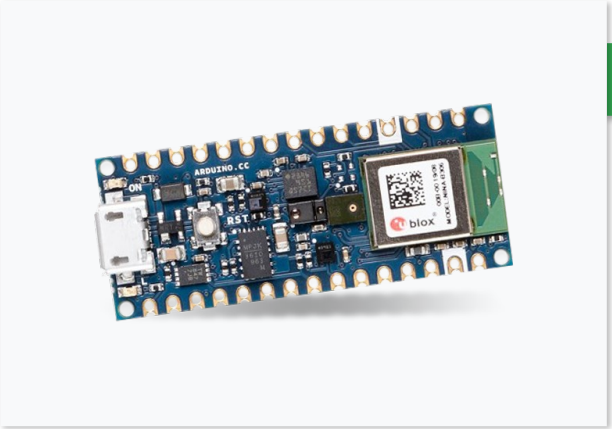
↓

TensorFlow Lite **Model File** *(.tflite)*

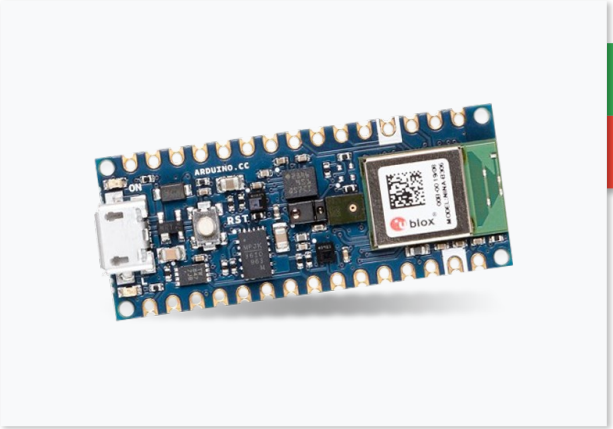
iOS


ARDUINO.CC
ON
RST
blox

?

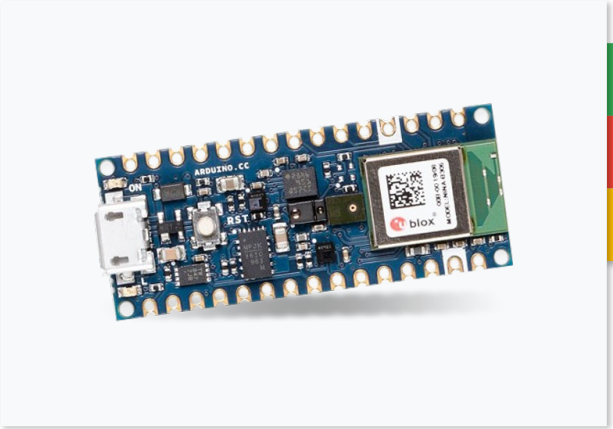


Even less memory



Even less memory

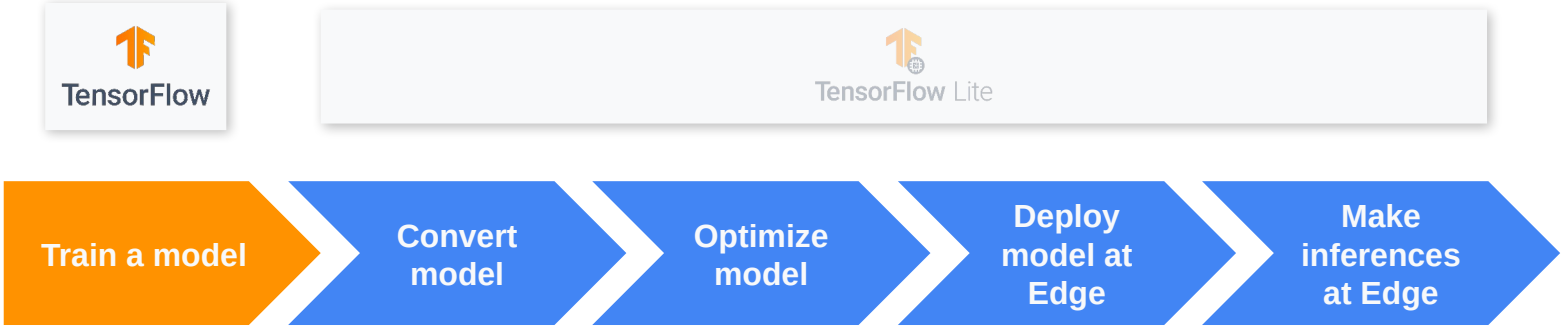
Even less compute power

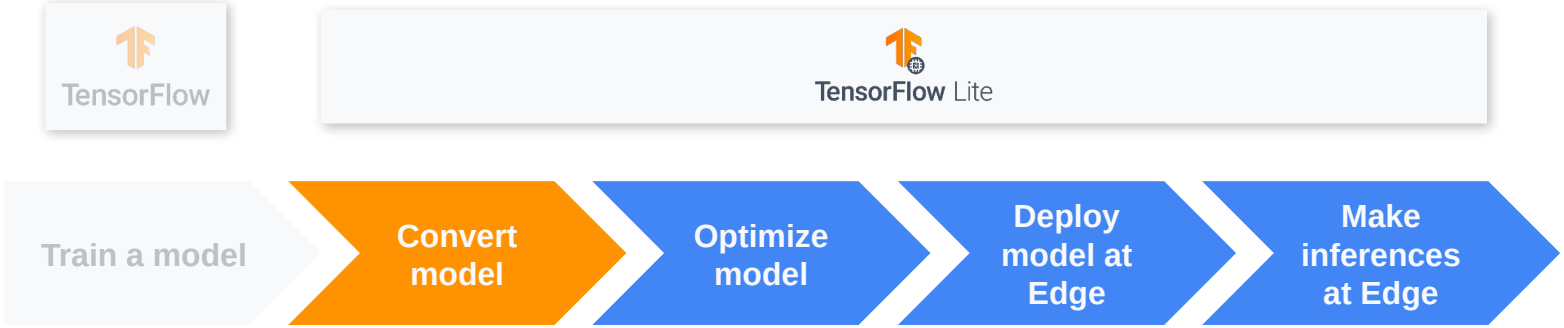


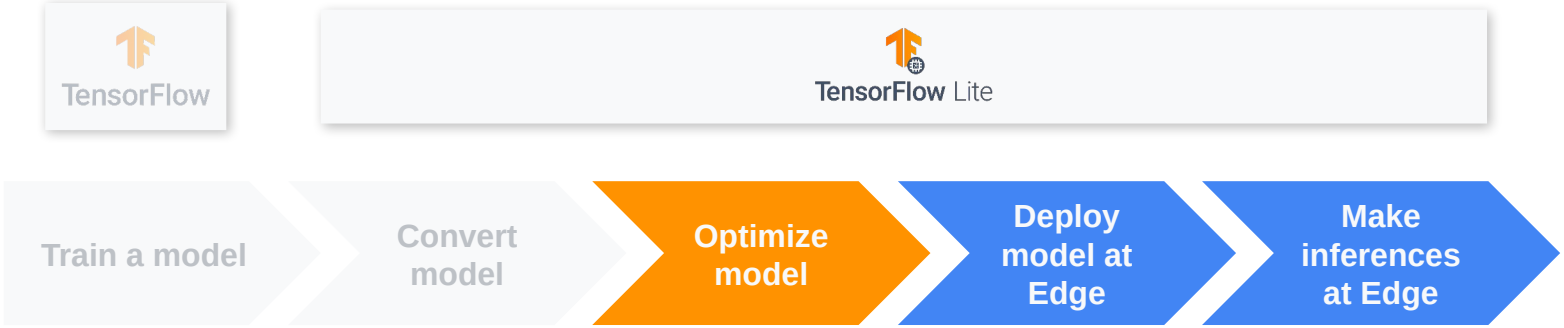
Even less memory

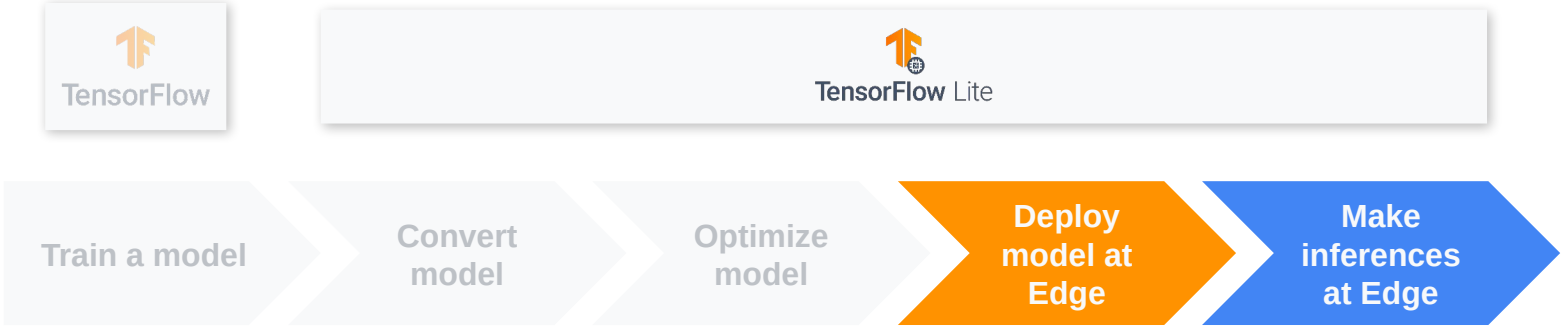
Even less compute power

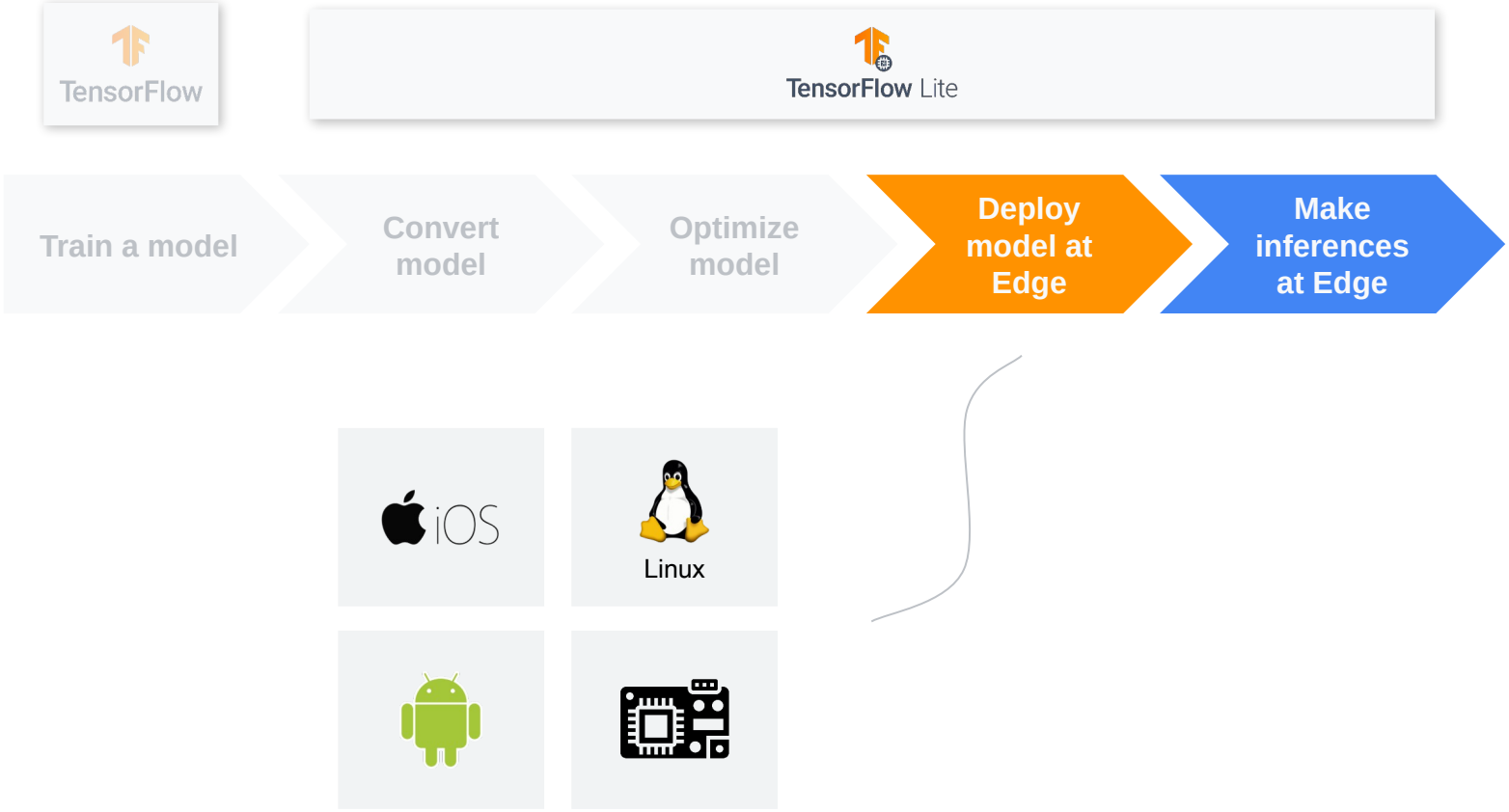
Also, only focused on ***inference***


TensorFlow
TensorFlow Lite
Train a model
Convert model
Optimize model
Deploy model at Edge
Make inferences at Edge


TensorFlow
TensorFlow Lite
Train a model
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TensorFlow
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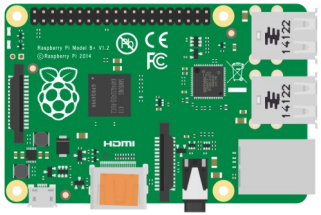

TensorFlow
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TensorFlow
TensorFlow Lite
Train a model
Convert model
Optimize model
Deploy model at Edge
Make inferences at Edge
iOS
Linux

TensorFlow

TensorFlow Lite

Train a model → Convert model → Optimize model → **Deploy model at Edge** → **Make inferences at Edge**



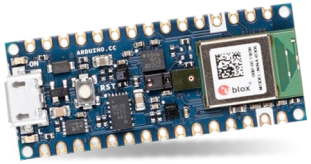
Raspberry Pi

Linux

TensorFlow

TensorFlow Lite

Train a model → Convert model → Optimize model → **Deploy model at Edge** → **Make inferences at Edge**



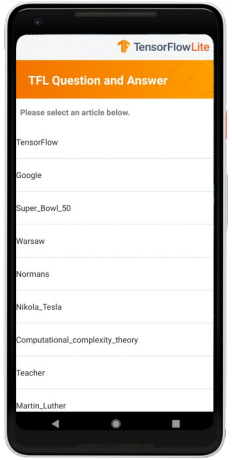
Microcontroller

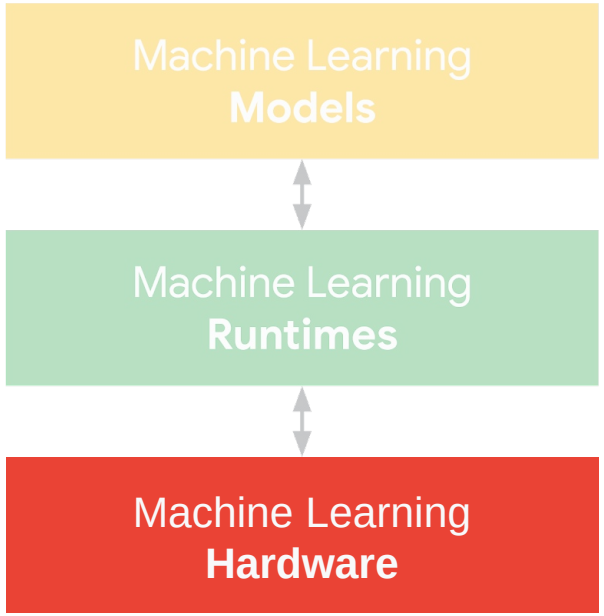
Linux

TensorFlow

TensorFlow Lite

Train a model → Convert model → Optimize model → Deploy model at Edge → **Make inferences at Edge**





Half-screen. Show presenter.

Fullscreen. Show presenter.